

UNIT 2: Cascading Style Sheets(CSS)

What are Cascading Style Sheets?

Cascading Style Sheets (CSS) are *rules*.

CSS are a way to control the look and feel of your HTML documents in an organized and efficient manner. CSS defines HOW HTML elements are to be displayed.

Each rule consists of a *selector* and a *declaration* (which, in turn, is made up of a *property* and a *value*).

They were established by the World Wide Web Consortium (W3C).

CSS rules control the look (*Style*) of web pages or XML files by providing central locations (*Sheets*) where HTML or XML tags are interpreted by the browser.

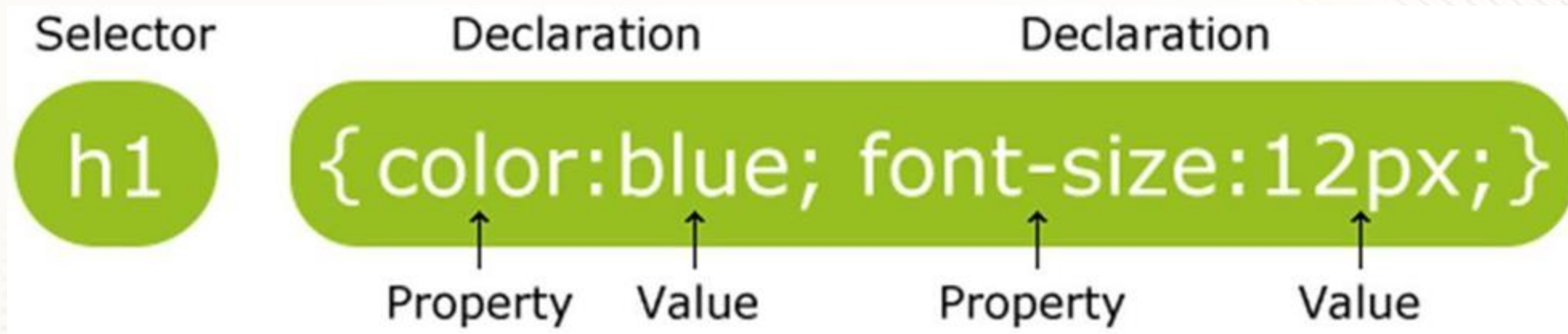
CSS Parts

A CSS rule has two main parts: **a selector**, and **one or more declarations**:

- ☐ The selector is normally the HTML element you want to style.
- ☐ Each declaration consists of a property and a value.
- ☐ The property is the style attribute you want to change.

Each property has a value.

A CSS declaration always ends with a semicolon, and declaration groups are surrounded by curly brackets



```
p    {  
    color: blue;  
    font-size: 20px;  
    width: 200px;  
}  
h1   {  
    color: green;  
    font-size: 36px;  
    text-align: center;  
}  
...  
```

Stylesheet

There are three ways of inserting a style sheet:

External CSS- It allows you to maintain a consistent look and feel across multiple pages of a website by reusing the same CSS file

Internal CSS-It's useful when you want to apply unique styles to a single HTML document

Inline CSS-It's useful when you need to apply unique or temporary styles to specific elements.

Inline CSS contains the CSS property in the body section attached with element. This kind of style is specified within an HTML tag using the *style attribute*. The style attribute can contain any CSS property.

```
<p style="color:red;  
margin-left:20px">  
This is a paragraph.</ p>
```

An internal style sheet should be used when a single document has a unique style. You define internal styles in the head section of an HTML page, by using the *<style>* tag, like this:

```
<head>  
<style type="text/ css">  
hr { color:red;}  
p {margin-left:20px;}  
body {background image:url("images/ back40.gif");}  
</ style>  
</ head>
```


An external style sheet is ideal when the style is applied to many pages. With an external style sheet, you can change the look of an entire Web site by changing one file. Each page must link to the style sheet using the <link> tag.

<head>

<link rel=" stylesheet" type="text/ css" href="mystyle.css" />

</ head>

An external style sheet can be written in any text editor. The file should not contain any html tags. Your style sheet should be saved with a .css extension.

hr{ color:sienna;}

p {margin-left:20px;}

body {background-image:url("images/ back40.gif");}

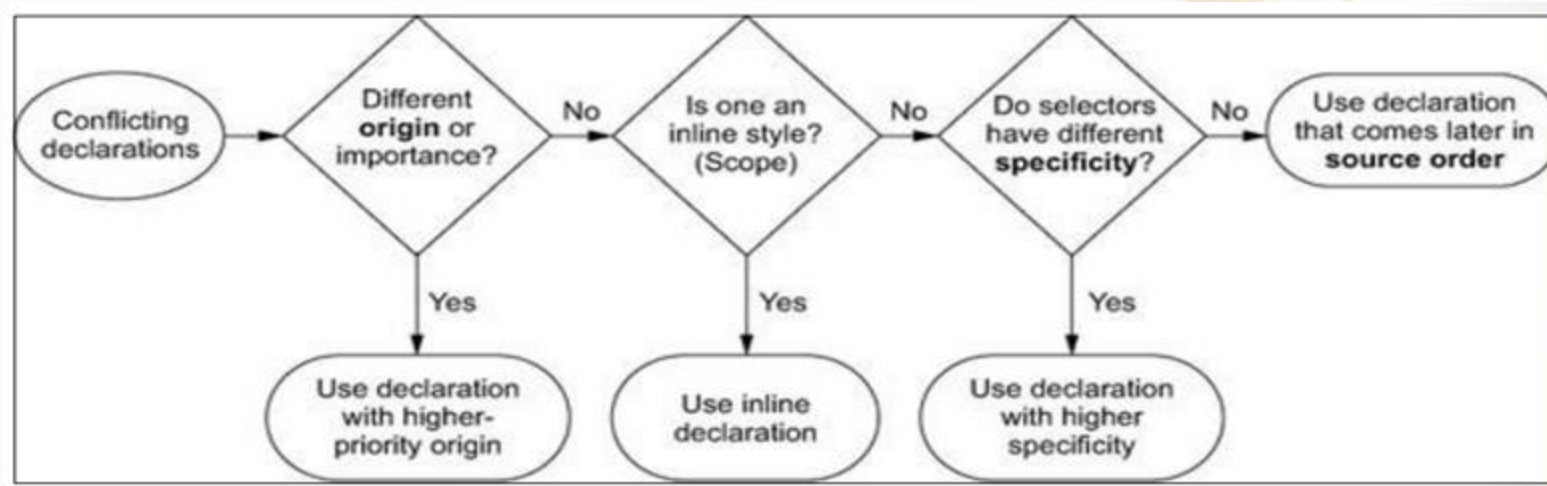
Cascading Order

How the cascade defines which CSS rules apply when more than one thing could style an element.

- Browser default
- External style sheet
- Internal style sheet (in the head section)
- Inline style (inside an HTML element)

an inline style (inside an HTML element) has the highest priority, which means that it will override a style defined inside the `<head>` tag, or in an external style sheet, or in a browser (a default value).

Note: If the link to the external style sheet is placed after the internal style sheet in HTML `<head>`, the external style sheet will override the internal style sheet!



When declarations conflict, the cascade considers three things to resolve the difference:

Stylesheet origin—Where the styles come from. Your styles are applied in conjunction with the browser's default styles.

Selector specificity—Which selectors take precedence over which.

Source order—Order in which styles are declared in the stylesheet.

Source order - Order in which styles are declared in the stylesheet.

```
h1 {  
  color: red;  
}  
h1 {  
  color: purple;  
}
```

<h1>This is my heading.</h1>

Output:

This is my heading.

An element selector is less specific — it will select all elements of that type that appear on a page - so will get a lower score

Selector specificity—Which selectors take precedence over which.

```
.main-heading {  
  color: red;  
}
```

```
h1 {  
  color: blue;  
}
```

```
<h1 class="main-heading">This is my heading.</h1>
```

Output:

This is my heading.

A class selector is more specific — it will select only the elements on a page that have a specific class attribute value — so will get a higher score.

Specificity:

Different types of selectors also have different specificities.

- An ID selector has a higher specificity than a class selector.
- Similarly, a class selector has a higher specificity than a element selector

If the two rulesets have the same specificity, the later one wins.

If they have different specificity, the one with the higher specificity wins.

Inline styles win over any specificity.

CSS selector

- A CSS selector selects the HTML element(s) you want to style.
- **Simple/Element type selectors** (select elements based on name)
- **ID Selector** (The id selector is used to specify a style for a single, unique element)
- **Class Selector** (It is used to specify a style for a group of elements)
- **Universal Selector** (The universal selector is used as a wildcard character. It selects all the elements on the pages)

Combinator selectors (select elements based on a specific relationship between them)

Pseudo-class selectors (select elements based on a certain state)

Pseudo-elements selectors (select and style a part of an element)

Attribute selectors (select elements based on an attribute or attribute value)

Element Selector

Element name

```
p {  
  color: blue;  
}
```

Blue text

```
p {  
  color: blue;  
}
```

```
...  
<p>...</p>  
...  
<p>...</p>
```

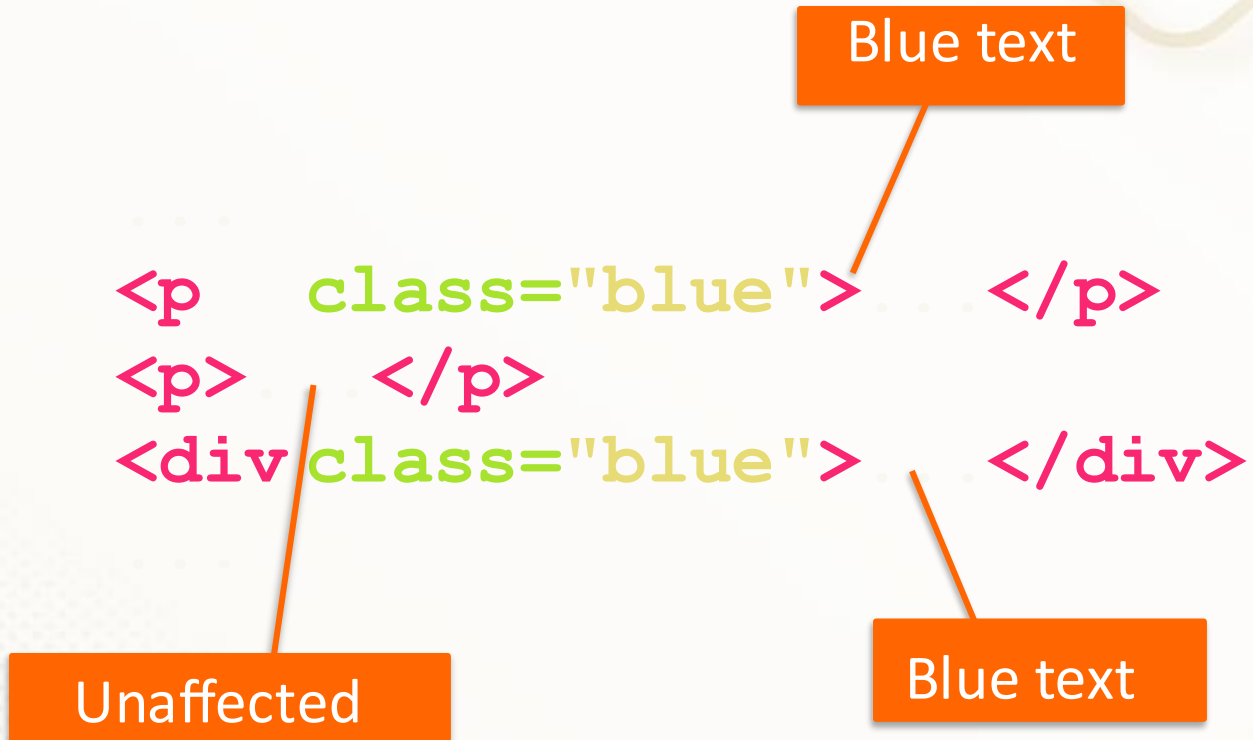
Blue text

class selector

Class name

Defined with .

```
.blue {  
    color: blue;  
}
```



id selector

id Value

```
#name {  
    color: blue;  
}
```

...
<p> . . . </p>
<div id="name"> . . . </div>
...

Unaffected

Blue text

Comparisons

Aspect	Element Selector	Class Selector	ID Selector
Syntax	Element/Tag name (e.g., p)	Dot followed by a class name (e.g., .my-class)	Pound/hash symbol followed by an ID name (e.g., #my-id)
Specificity	Low specificity - targets all elements of the specified type	Moderate specificity - targets elements with a specific class	High specificity - targets a single element with a specific ID
Use Case	Styling all paragraphs (<p>) in a document	Styling multiple elements with a common class, e.g., navigation links	Styling a specific, unique element such as a header
HTML Example	This is a paragraph.	Link	<div id="my-id">Unique Div</div>

Styles can be grouped:

- Using
 - multiple styles
 - multiple selectors
 - contextual selectors
 - direct child selectors
 - adjacent selectors
- By attribute

A combinator is something that explains the relationship between the selectors.

There are four different combinators in CSS:

- Descendant Selector (space)
- Child Selector ($>$)
- Adjacent Sibling Selector ($+$)
- General Sibling Selector ($\sim$) - It selects all **next** elements that are siblings of a specified element.

Let us see how we can use different symbols ($+$, $>$, and $\sim$) in CSS selectors and their differences.

Each rule can include *multiple styles* using semicolons to separate them:

```
h2 { color: darkblue;  
      font-style: italic;}
```

Additionally, *multiple selectors* that have the same styles can be grouped using commas to separate them:

```
h1, h2, h3 { color: yellow; }
```

Separate selectors
with commas

```
div, .blue  
{ color:  
  blue;  
}
```

Blue text

```
...  
<p class="blue">...</p>  
<p>...</p>  
<div>...</div>  
...
```

Blue text

Combining Selectors

Every **p** that has

- Element With Class Selector

```
p.big {  
  font-size:  
  20px;  
}
```

NOTE lack of space between element and class definition

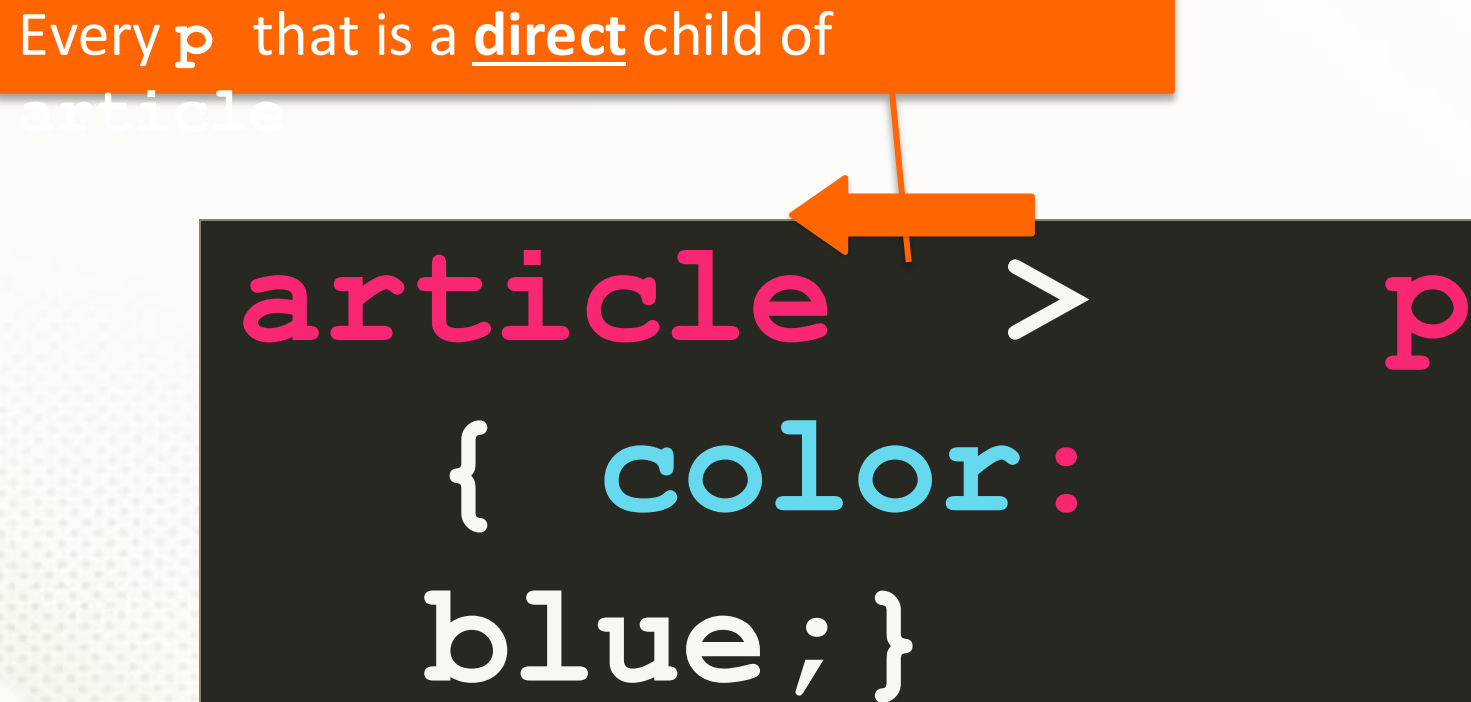
Text size 20px

```
...  
<p class="big">...</p>  
<div class="big">...</div>  
...
```

Unaffected text

Child Selector

Every **p** that is a direct child of
article



```
article > p  
{ color:  
blue; }
```

Blue text

```
<article>...  
  <p>...</p>  
</article>
```

Unaffected text

```
...  
<p>...</p>  
<article>...  
  <div><p>...</p></div>  
</article>
```

Unaffected text

Descendant Selector

Every **p** that is inside (at any level) of
article



```
article p{  
color : blue;  
}
```

Blue text

```
<article>...  
  <p>...</p>  
</article>
```

Unaffected text

```
...  
<p>...</p>  
<article>...  
  <div><p>...</p></div>  
</article>
```

Blue text


```
.colored p {  
  color: blue;  
}
```

Every **p** that is inside (at any level) an element with **class="colored"**

```
article >  
  .colored {  
    color: blue;  
  }
```

Every element with **class="colored"** that is a direct child of **article** element

Pseudo-classes

A pseudo-class is used to define a special state of an element.

For example, it can be used to:

1. Style an element when a user mouses over it
2. Style visited and unvisited links differently
3. Style an element when it gets focus

```
selector:pseudo-class {  
    ...  
}
```

Pseudo-elements

A CSS pseudo-element is used to style specified parts of an element.

For example, it can be used to:

1. Style the first letter, or line, of an element
2. Insert content before, or after, the content of an element

```
Selector::pseudo-element {  
  property: value;  
}
```

CSS properties

Color and Text Properties:

- `color`: Sets the text color.
- `font-family`: Specifies the font family for text.
- `font-size`: Sets the font size.
- `font-weight`: Controls the thickness of the characters.
- `text-align`: Aligns text within its container.
- `text-decoration`: Sets text decorations like underline, overline

Background Properties

- background-color: Sets the background color of an element.
- background-image: Specifies an image to use as the background.
- background-repeat: Controls how a background image repeats.
- background-size: Sets the size of the background image.

Background - Shorthand property

To shorten the code, it is also possible to specify all the properties in one single property. This is called a shorthand property.

```
body {background:#ffffff url("img_tree.png") no-repeat right top;}
```

When using the shorthand property the order of the property values are:

background-color

background-image

background-repeat

background-attachment

background-position

It does not matter if one of the property values is missing, as long as the ones that are present are in this order.

Border Properties

The CSS border properties allow you to specify the style, width and color of an element's border

The **border-style** property specifies what kind of border to display.

The following values are allowed:

- dotted - Defines a dotted border
- dashed - Defines a dashed border
- solid - Defines a solid border
- double - Defines a double border
- groove - Defines a 3D grooved border. The effect depends on the border color value
- ridge - Defines a 3D ridged border. The effect depends on the border-color value
- inset - Defines a 3D inset border. The effect depends on the border-color value
- outset - Defines a 3D outset border. The effect depends on the border-color value
- none - Defines no border
- hidden - Defines a hidden border

Border - Shorthand Property

The border property is a shorthand property for the following individual border properties:

- border-width
- border-style (required)
- border-color

TextProperties

Text Color- The color property is used to set the color of the text.

- `body { color:blue; }`
- `h1 { color:#00ff00; }`
- `h2 { color:rgb(255,0,0); }`

☐ **Text Alignment**

- ☐ The text-align property is used to set the horizontal alignment of a text.
- ☐ Text can be centered, or aligned to the left or right, or justified.
- ☐ When text-align is set to "justify", each line is stretched so that every line has equal width, and the left and right margins are straight (like in magazines and newspapers)

`h1 {text-align:center; }`

`p.main {text-align:justify; }`

Text Decoration

The text-decoration property is used to set or remove decorations from text.

- `h1 {text-decoration:overline;}`
- `h2 {text-decoration:line-through;}`
- `h3 {text-decoration:underline;}`
- `a {text-decoration:none;}` remove underlines from links

Text Transformation

It can be used to turn everything into uppercase or lowercase letters, or capitalize the first letter of each word.

```
p {text-transform:uppercase;}
```

Text Indentation

It is used to specify the indentation of the first line of a text.

```
p {text-indent:50px;}
```

Word Spacing

It is used to specify the exact value of the spacing between your words.

```
p { word-spacing: 10px; }
```

Letter Spacing

It is used to specify the exact value of the spacing between your letters.

```
p { letter-spacing: 3px; }
```

Line Height

The line-height property is used to specify the space between lines

```
p {line-height:1.8}
```

Text direction

The direction property is used to change the text direction of an element:

```
p { direction:rtl;}
```

Font Properties

Order must be: style, variant weight, size, line height,font family(s)

p{font:font-style font-variant font-weight font-size/ line -height font-family}

Font Family

If the name of a font family is more than one word, it must be in quotation marks, like font-family: "Times New Roman".

More than one font family is specified in a comma-separated list:

p{font-family:"Times New Roman",";}

Font Style

This property has three values:

- normal - The text is shown normally
- italic - The text is shown in italics
- oblique - The text is "leaning" (oblique is very similar to italic, but less supported)

□ **p{font-style:normal;}**

Font Size - sets the size of the text. `p {font-size:14px;}`

`p {font-size:0.875em;} / * 14px/ 16=0.875em */ p { font-size: 20%; }`

- The default text size in browsers is 16px. So, the default size of 1em is 16px.

- **Font Color**

- `h4 { color: red; }`

- **Font Weight**

If you want to control the weight of your font (its thickness), using font weight is the best way to go about it.

You only use font-weight in multiples of 100 (e.g. 200, 300, etc) .The values range from 100 (thin)-900 (thick).

- `p { font-weight: 100; }`

- `ul{ font-weight: bolder; }`

Available key terms for font-weight: ***bold or bolder, lighter and normal.***

Font Variant

- allows you to convert your font to all lower case to upper case with small size font.

```
p { font-variant: small-caps; }
```

```
p { font-variant: normal; }
```


Table styles

- **table:** This selector targets all `<table>` elements in your HTML document.
- **width: 80%:** It sets the width of the table to 80% of its containing element's width. This makes the table responsive and not span the entire width.
- **margin: 20px auto:** It adds a margin of 20 pixels on all sides, and auto horizontally centers the table within its containing element.
- **border-collapse: collapse:** This property ensures that the borders of adjacent table cells (both internal and external) are collapsed into a single border. It gives a cleaner look to the table.
- **border: 2px solid #333:** This adds a 2-pixel wide solid border with a color of #333 (a shade of gray) around the entire table.

- `caption`: This selector targets the `<caption>` element within the table.
- `th`: This selector targets all table header cells (`<th>` elements) in the table.
- `td`: This selector targets all table data cells (`<td>` elements) in the table.
- `tbody tr:nth-child(odd)`: This selector targets all odd-numbered rows within the `<tbody>` section of the table.
- `tbody tr:hover`: This selector targets any row within the `<tbody>` section of the table when the user hovers the mouse pointer over it.

The display Property

- The display property specifies if/how an element is displayed.
- Every HTML element has a default display value depending on what type of element it is. The default display value for most elements is block or inline.

CSS Opacity / Transparency

- In CSS, you can control the opacity or transparency of an element using the opacity property. The opacity property allows you to specify how transparent or opaque an element and its content should be. It accepts values between 0 (completely transparent) and 1 (completely opaque), where decimal values represent varying levels of transparency.